

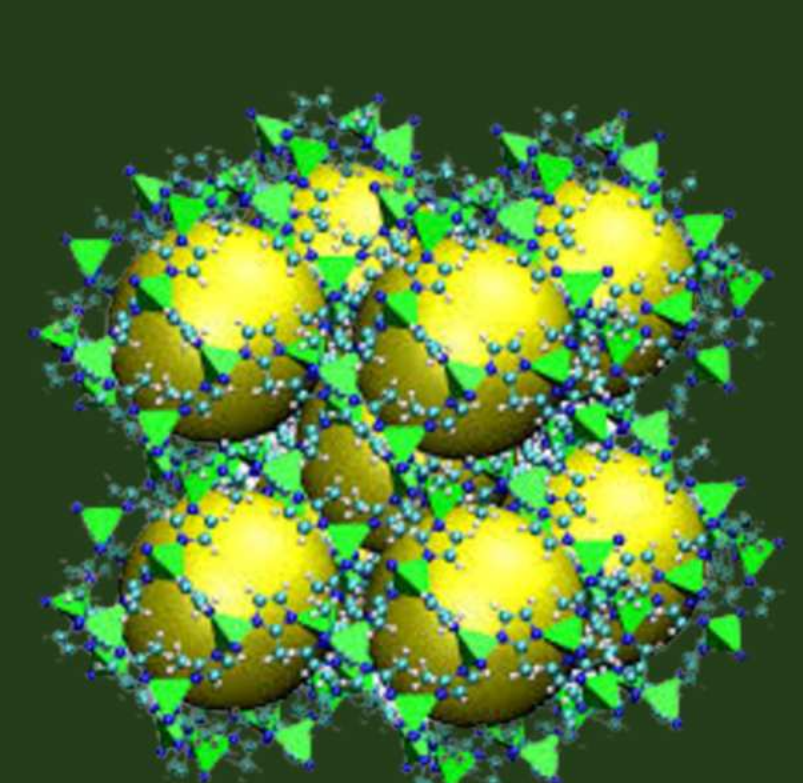


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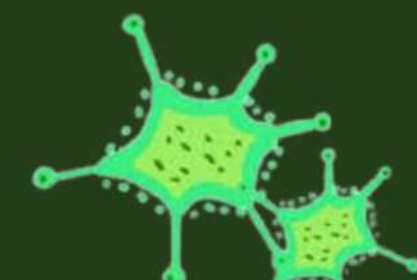


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CRYSTALCARE: OPTIMIZED NANOCRYSTALS FOR DRUG DELIVERY

Discovering Optimized Nanocrystal Framework Drug Delivery Systems For Diversified Delivery Applications

Biomedical Engineering | BMED

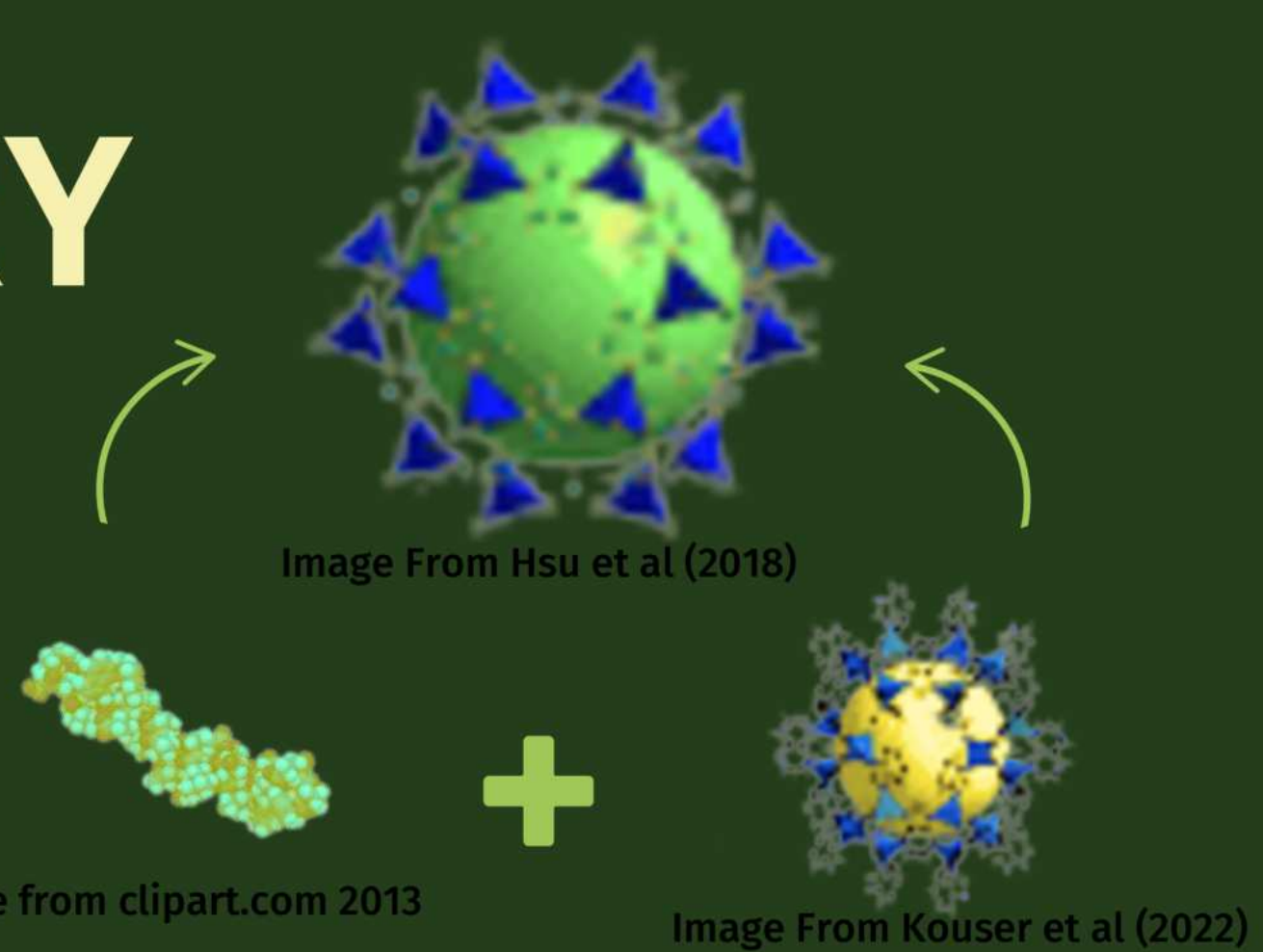
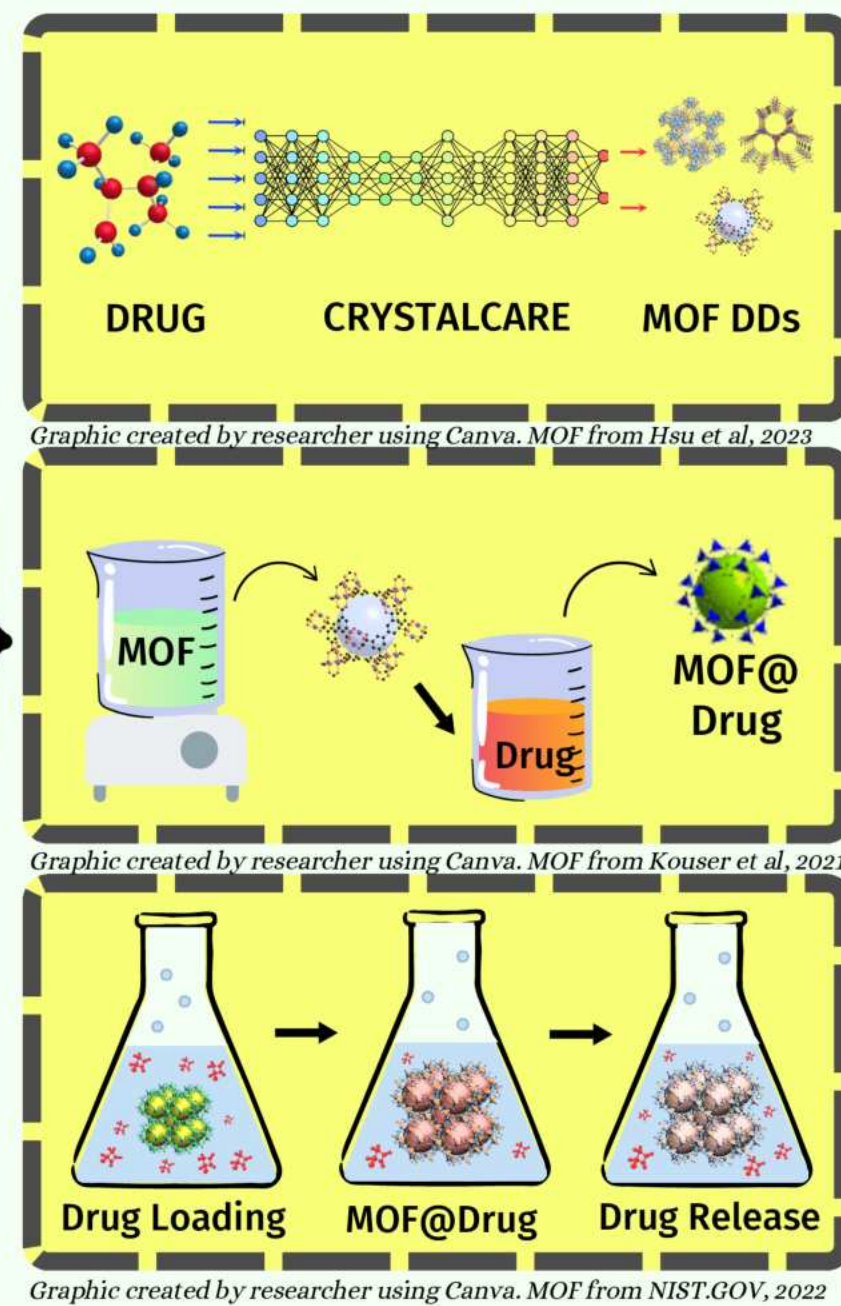


Image from clipart.com 2013

Image From Kouser et al (2022)

BRIEF OVERVIEW

This year's research aimed to engineer a **graph attention-based computational architecture** to identify & optimize Metal-Organic Frameworks as drug delivery systems. **Top-ranked MOFs were synthesized** via a UARM method and experimentally compared to standard-of-care drugs. The **discovered MOFs exhibited a significant ($p < 0.01$) increase in drug loading** when compared to leading pharmaceutical products. The CRYSTALCARE architecture presents a striking framework for pharmacological innovation.



INTRODUCTION

DRUG DELIVERY

Over 90% of viable drugs fail the FDA process, with over 50% of conventional drugs failing to reach their intended site of action due to **rapid degradation** and **poor bioavailability**. New approaches are required to overcome biological barriers and ensure that therapeutic agents are released precisely at the disease site. **Developing novel, biocompatible carrier systems** that provide targeted, controlled-release drug loading is an crucial area for **improving clinical outcomes in chronic diseases and infections**.

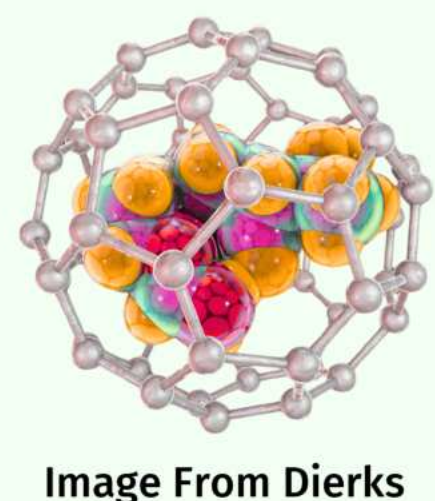


Image From Dierks (et al, 2017)

METAL ORGANIC FRAMEWORKS (MOFs)

MOFs are porous materials constructed from metal ions and organic linkers, known for **exceptional surface area**. They are utilized for high-capacity adsorption, antibacterial, and drug delivery applications. Further research on stabilizing these structures for therapeutic and biomedical use is needed.

HIGH SURFACE AREA

Maximizes high drug loading (7,000 m²/g) and antibacterial effectiveness

TUNABLE CHEMISTRY

Customizable structures enable selective antibacterial activity

DRUG DELIVERY

Enhances high drug stability, targeted release, and antibacterial effects

OBJECTIVES

This research hopes to redefine the current methods of drug delivery design. Instead of aiming to arbitrarily select drug delivery systems for experimental verification, this research aims to **design an optimization architecture that identifies MOF candidates with high potential as a drug delivery system** for an inputted drug, ultimately increasing ADME scores for drug use.

NOVEL ARCHITECTURE

Use attention based mechanisms to learn drug-MOF pairs.

TRANSPARENT RESULTS

Highlight valuable drug delivery MOF sites for researchers to analyze.

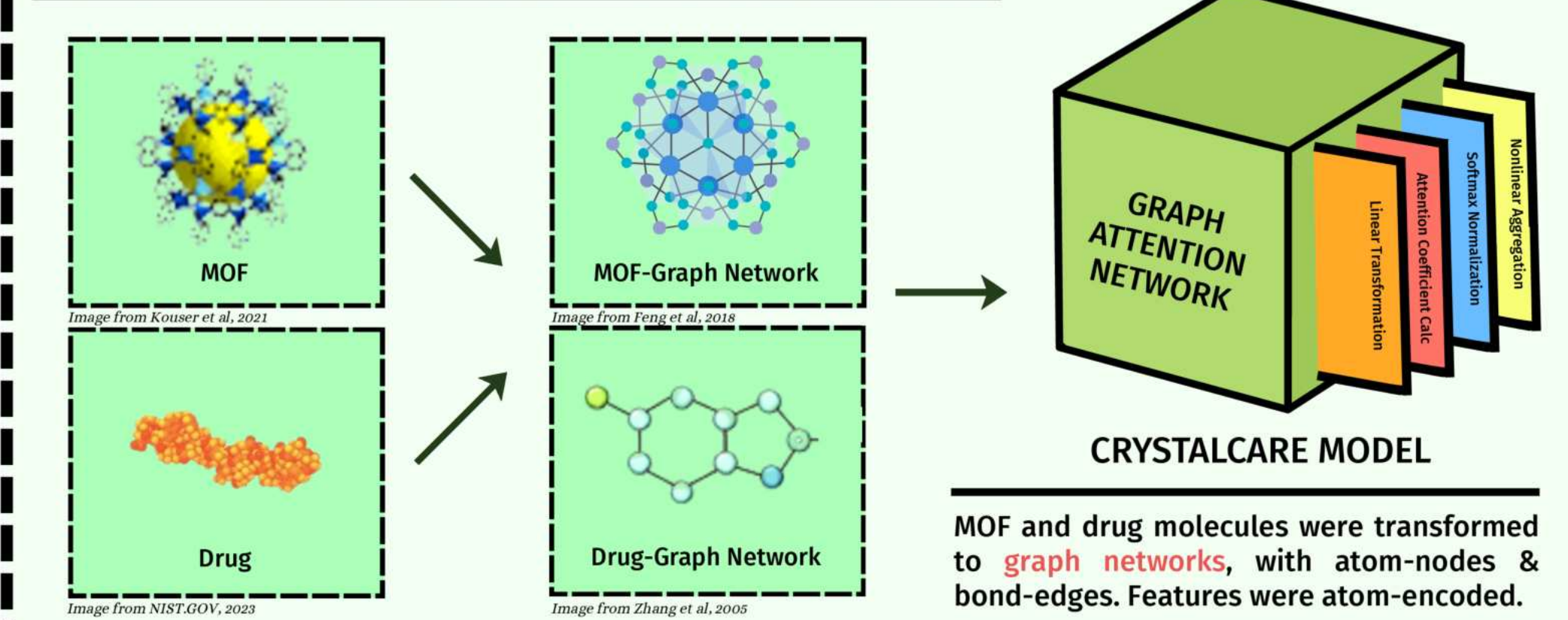
EXPERIMENTAL ANALYSIS

Experimentally synthesize and verify MOF DDs using drugs.

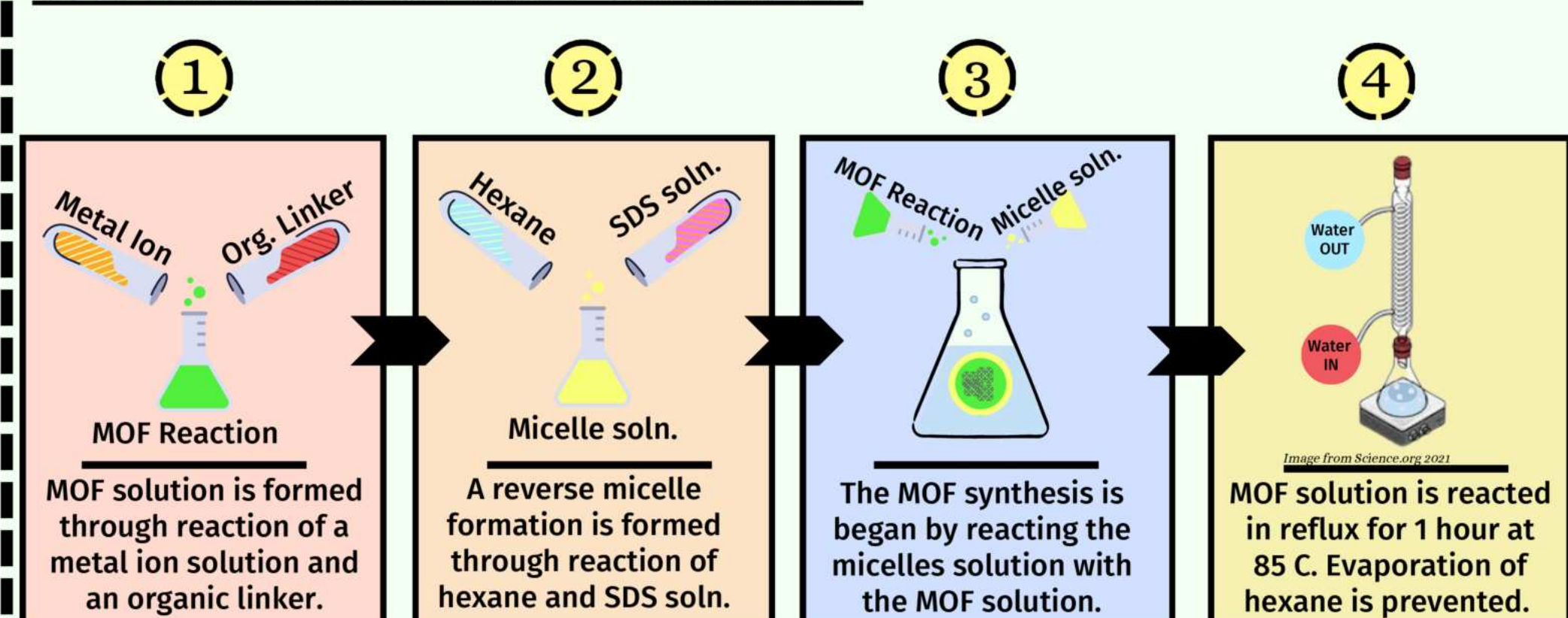
In addition, this research hopes to present new insights into drug loading and delivery in the field of chronic disease, a field that rarely has innovation in **drug optimization** and application. This research can hopefully provide insights into drug loading and possibility of new treatment mechanisms that can be applied to new applications, such as other sciences. **We hope CRYSTALCARE will save patients who lack approved drugs.**

METHODOLOGY

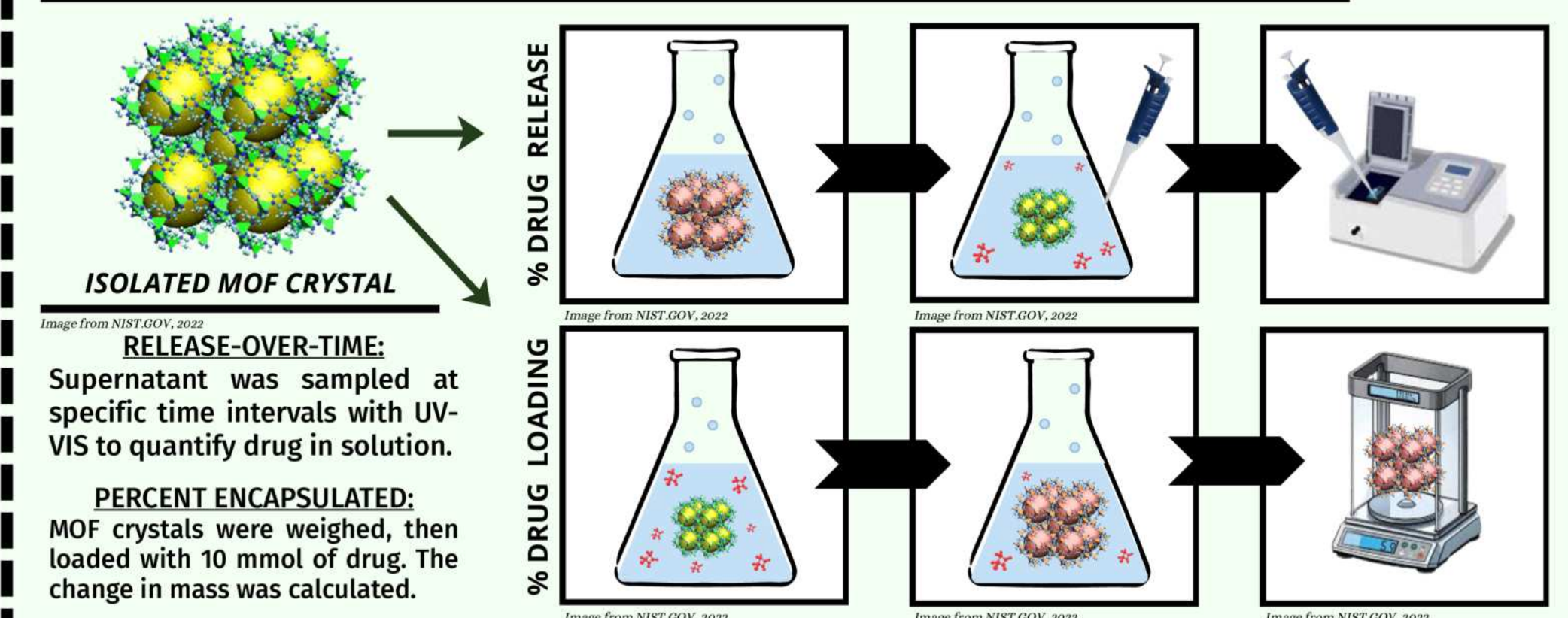
1. TRAINING OF CRYSTALCARE MODEL



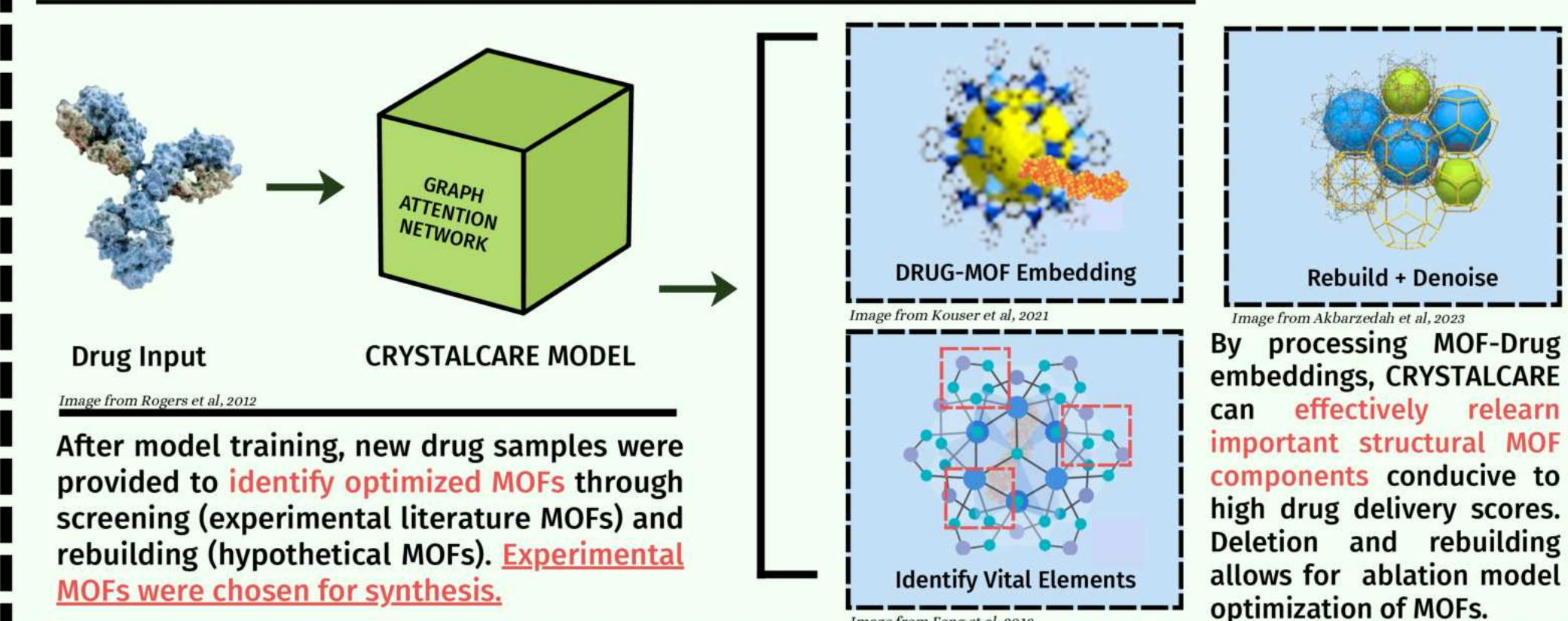
3. SYNTHESIS OF MOF CRYSTALS



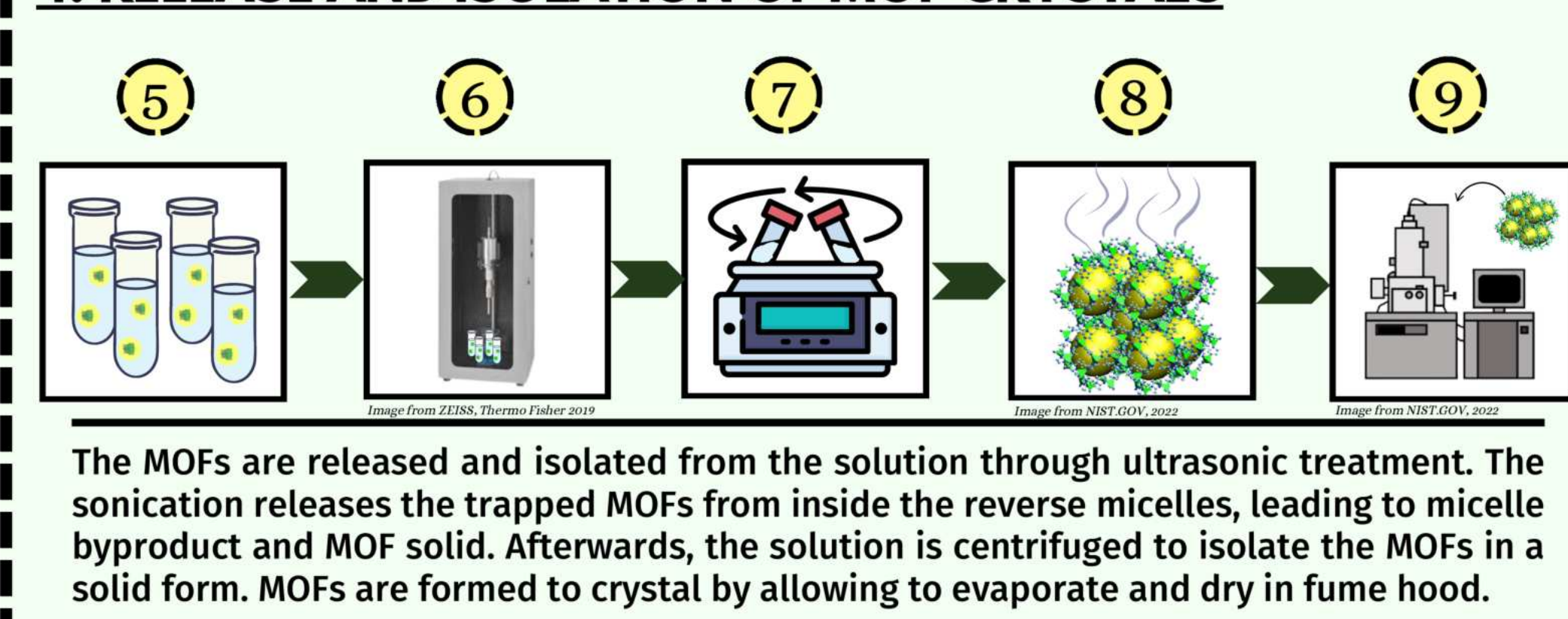
5. EXPERIMENTAL VERIFICATION OF DRUG DELIVERY



2. OPTIMIZATION OF MOF DELIVERY SYSTEMS



4. RELEASE AND ISOLATION OF MOF CRYSTALS

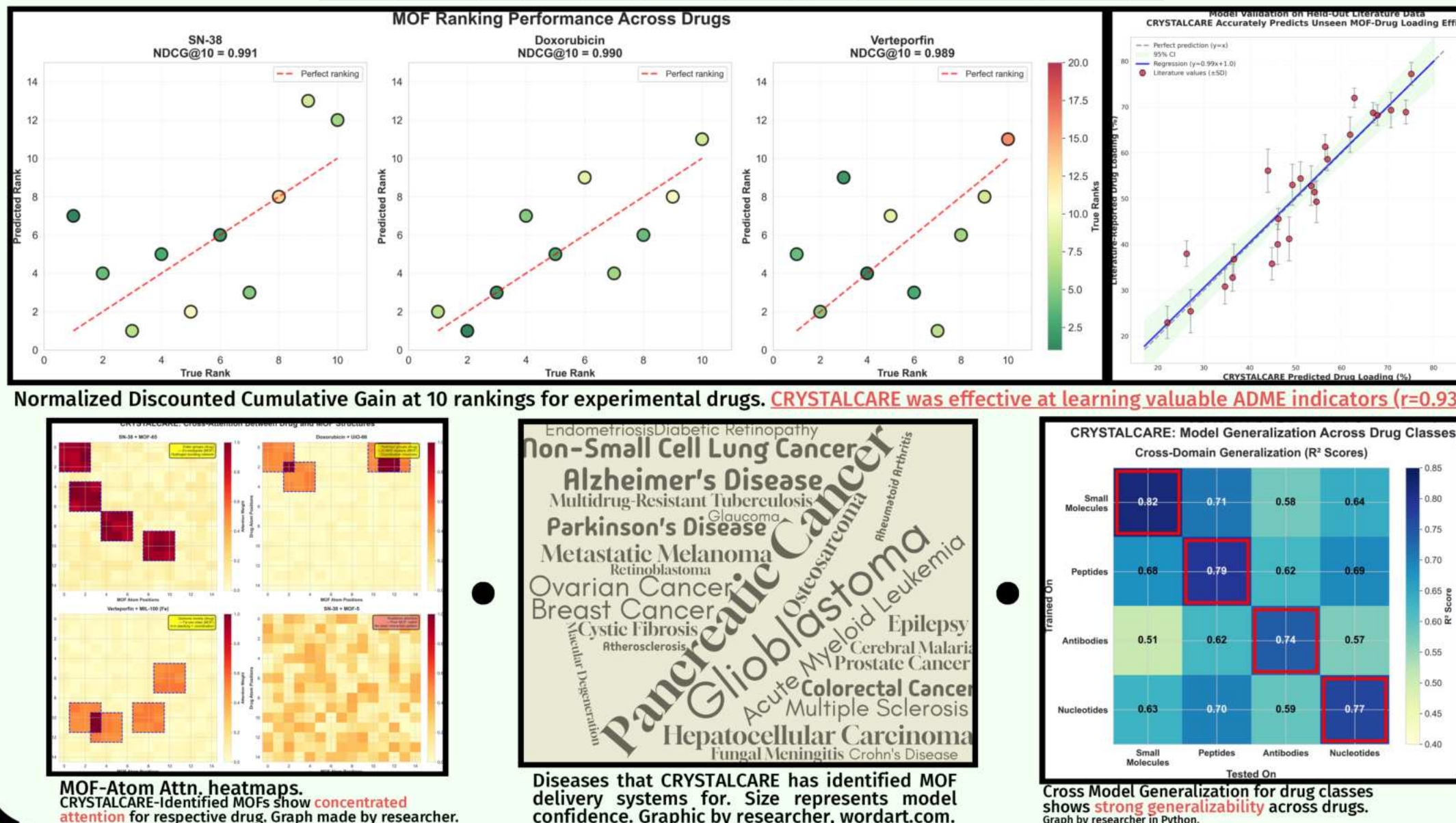


OVERVIEW OF METHODOLOGY

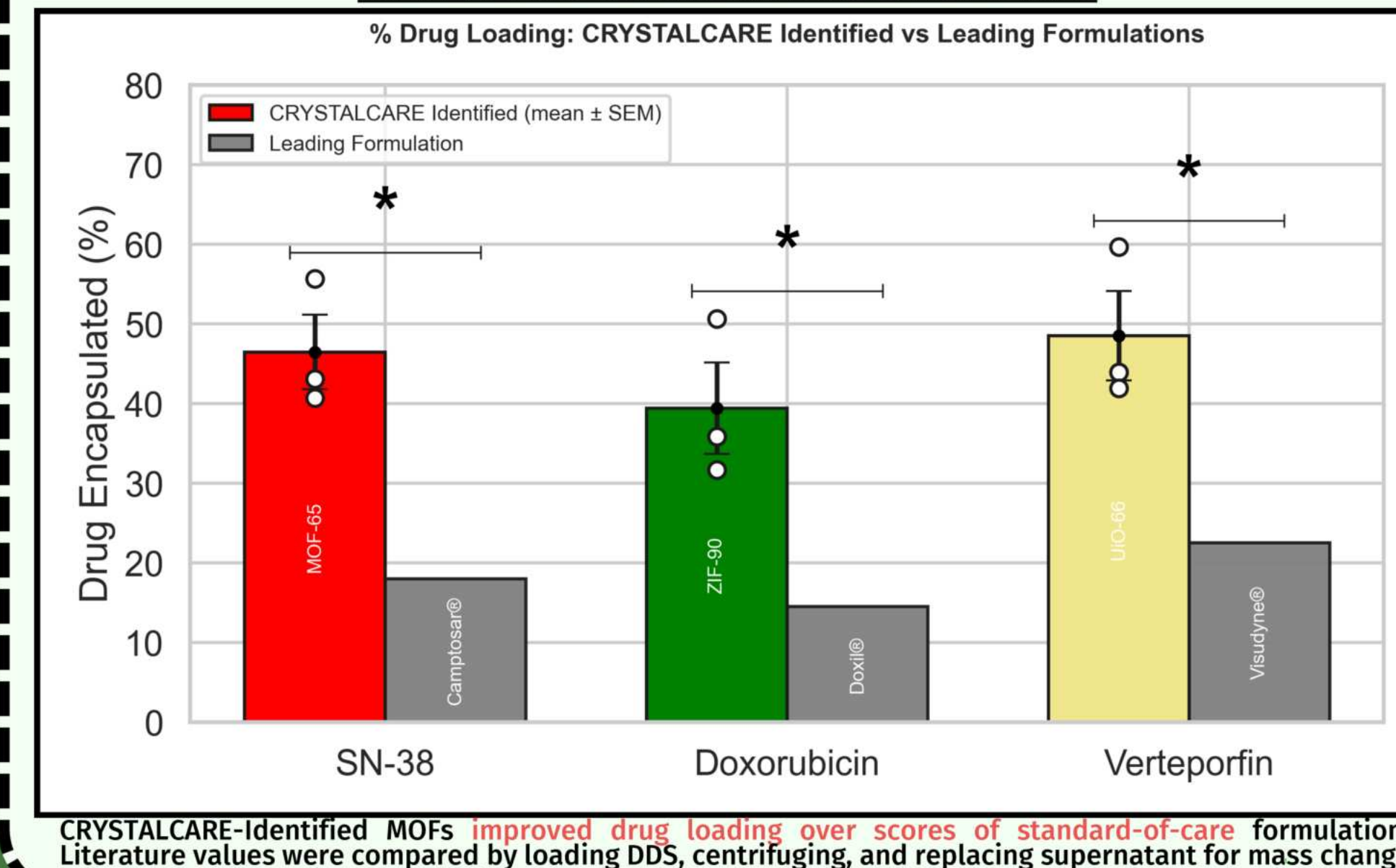
In this project, an **attention-based graph neural network** was trained by transforming MOF & drug molecules into graphical network representations. The model was used to **identify optimized MOF delivery systems for experimental validation on three selected drugs**. The three identified MOFs were synthesized via a UARM (ultrasonic-assisted reverse micelle) process and quantified through FTIR and SEM. MOFs were loaded with their respective drugs post-synthesis, whereby a **drug-loading assay** and **release-over-time assay** were completed. Drug release was quantified with UV-VIS spectroscopy curves.

RESULTS AND SYNTHESIS ANALYSIS

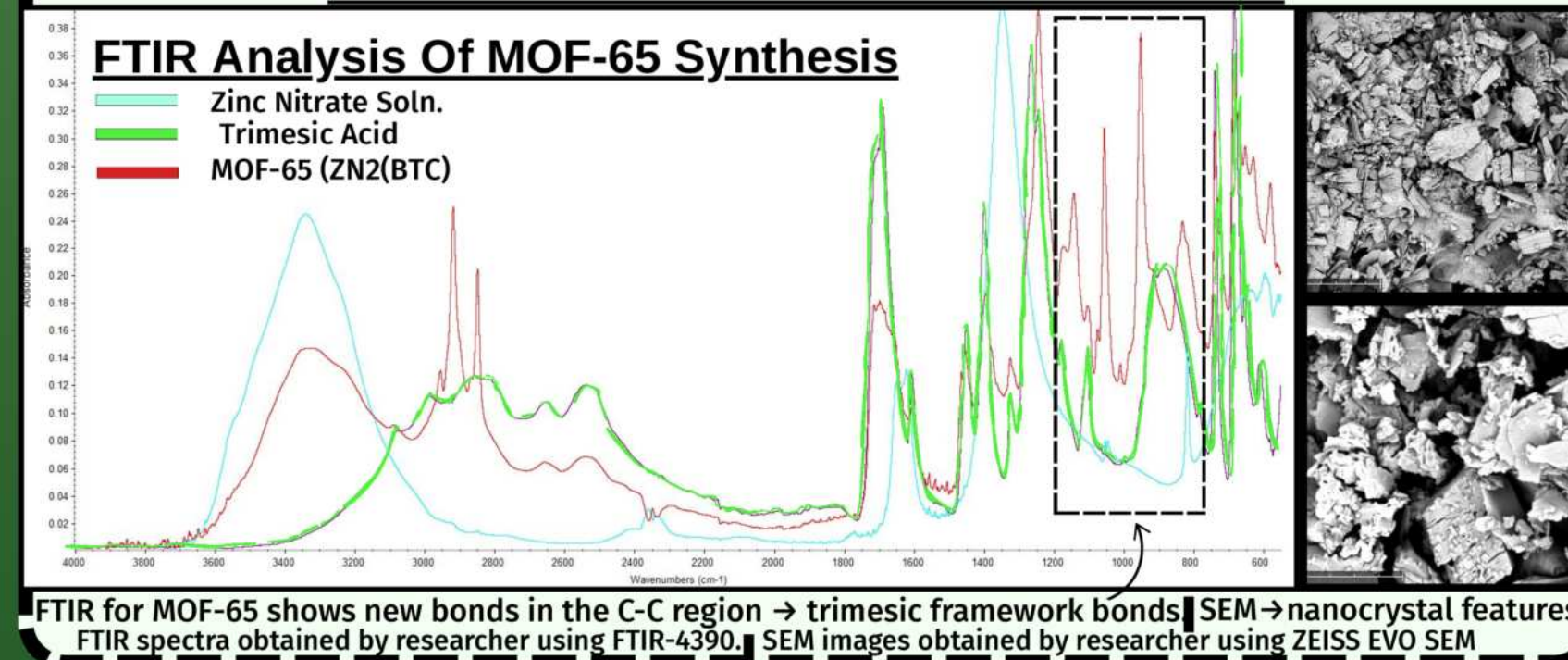
COMPUTATIONAL VALIDATION: CRYSTALCARE



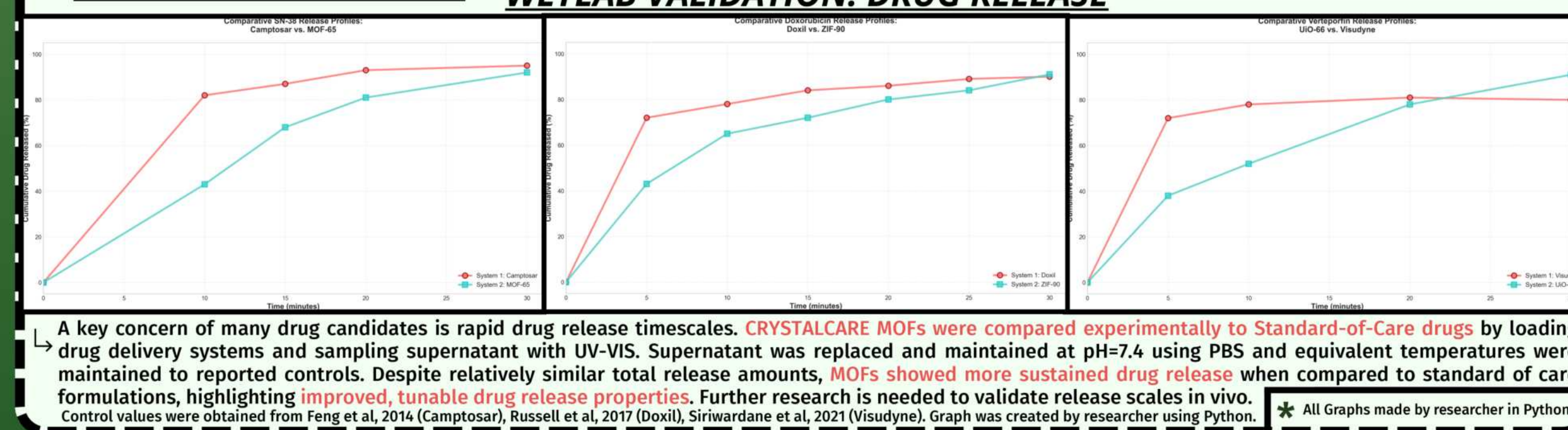
WETLAB VALIDATION: DRUG LOADING



WETLAB VALIDATION: MOF SYNTHESIS



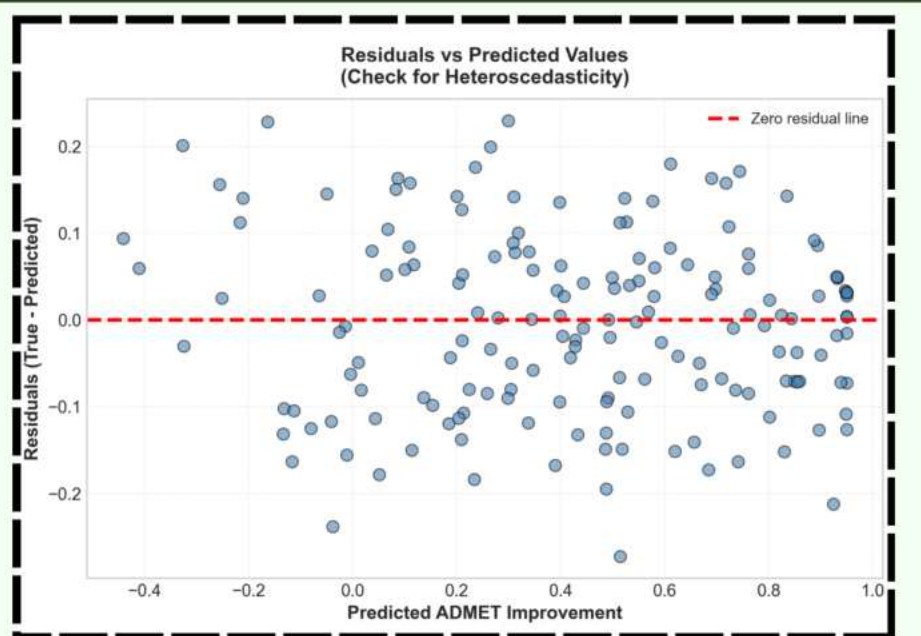
WETLAB VALIDATION: DRUG RELEASE



STATISTICAL ANALYSIS

A Breusch-Pagan test for homoscedasticity was run to analyze if residual variation was dependent on the independent variable. With $p=0.87$, we fail to reject the null hypothesis, suggesting that **residuals are independent**. A Shapiro-Wilk test was run to analyze normality of residuals; with $p < 0.01$, we reject the null hypothesis; **residuals are abnormally spread**.

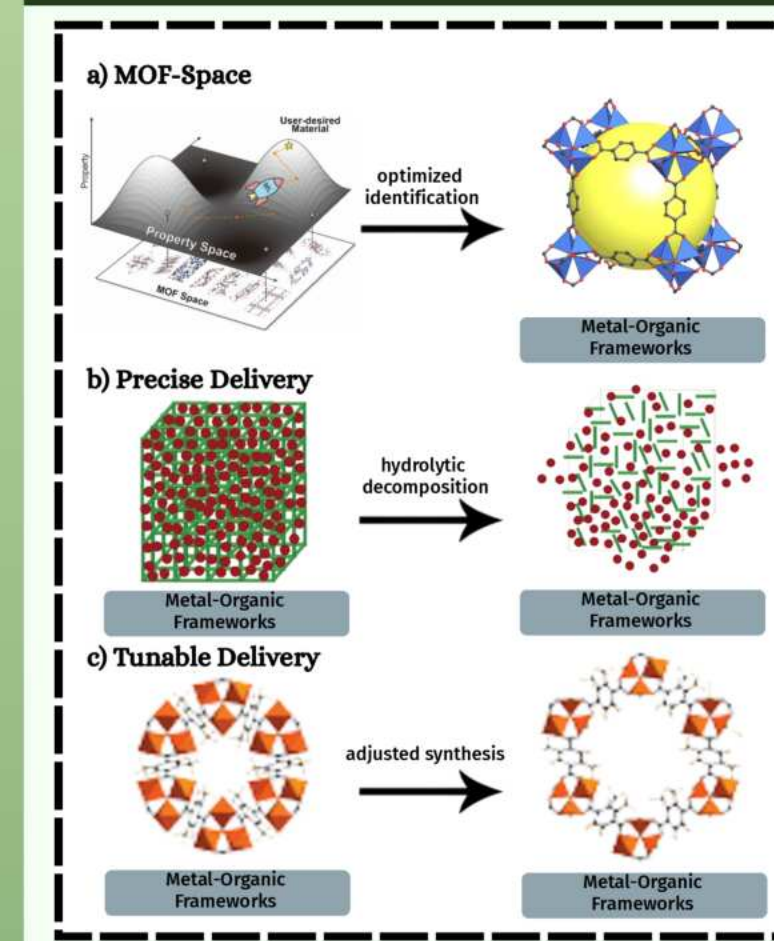
MOF DDs	Mean	SOC value	U	p value
MOF-65	47.453	18.756	0.00	0.02472
ZIF-90	39.928	14.392	0.01	0.02676
UIO-66	48.628	23.362	0.04	0.03245



Mann-Whitney U was used for drug loading significance in difference between MOFs and Standard-Of-Care. Null hypothesis was rejected.

CONCLUSIONS

MODEL VIABILITY



The CRYSTALCARE model **learned drug delivery patterns of MOF molecules**. This was validated using weight permutations and comparisons. CRYSTALCARE achieved 93% accuracy. The model showed signs of generalizability, highlighting **attention between MOF-node clusters at higher-order scales**. This model can empower drugs through identification of **optimal drug delivery systems, increasing the likelihood of drugs reaching FDA approval**.

MOF DRUG DELIVERY

This project **synthesized the top three identified MOFs from the CRYSTALCARE model**.

These MOFs were quantified using FTIR and SEM. MOFs **improved delivery** by drug loading over standard-of-care drugs ($p < 0.01$). MOF drug delivery systems showcase **significant promise** for future clinical research and testing.

MODEL NOVELTY

CRYSTALCARE used an attention head to learn local and global patterns

DRUG DELIVERY

Optimized MOFs showed high delivery potential through high drug loading.

POTENTIAL RESEARCH

CRYSTALCARE-identified MOFs can encapsulate drugs before testing.

FUTURE GOALS

1. The CRYSTALCARE model can be used to identify possible drug delivery systems for all failed drugs in the FDA system's history, **empowering millions of new treatments**.
2. MOF Drug delivery systems identified by CRYSTALCARE must be subjected to **rigorous in vivo testing** in both cell and mammalian models to ensure delivery predictions.
3. Further research into the **new designs of the hypothetical, optimal MOFs** provided by CRYSTALCARE could realign materials science researchers with chemical design goals.

REFERENCES

CANVA DESIGN WAS USED IN CONJUNCTION WITH CITED IMAGES UNLESS OTHERWISE CITED, IMAGES & GRAPHS BY RESEARCHER

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